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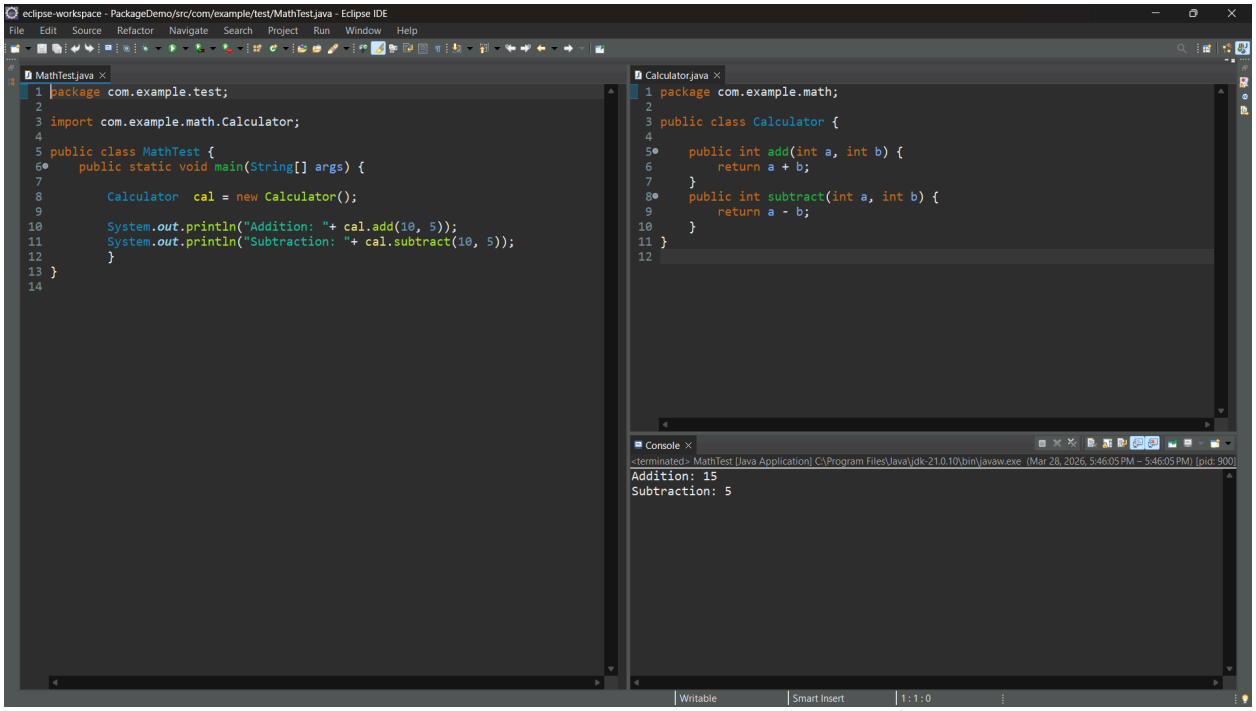
Subject: JAVA
Assignment: 7
Form Number: 251101107

Problem 1: Basic Package Creation and Usage
Task: Create a package named com.example.math and add a class Calculator with methods for addition and subtraction. Then, create another class, MathTest, in the com. example.test package to use the Calculator class

Solution:

```
package com.example.math;

public class Calculator {

	public int add(int a, int b) {

		return a + b;

	}

	public int subtract(int a, int b) {

		return a - b;

	}

}


package com.example.test;


import com.example.math.Calculator;


public class MathTest {

	public static void main(String[] args) {
```

```
        Calculator  cal = new Calculator();


        System.out.println("Addition: "+ cal.add(10, 5));

        System.out.println("Subtraction: "+ cal.subtract(10, 5));

        }

}
```



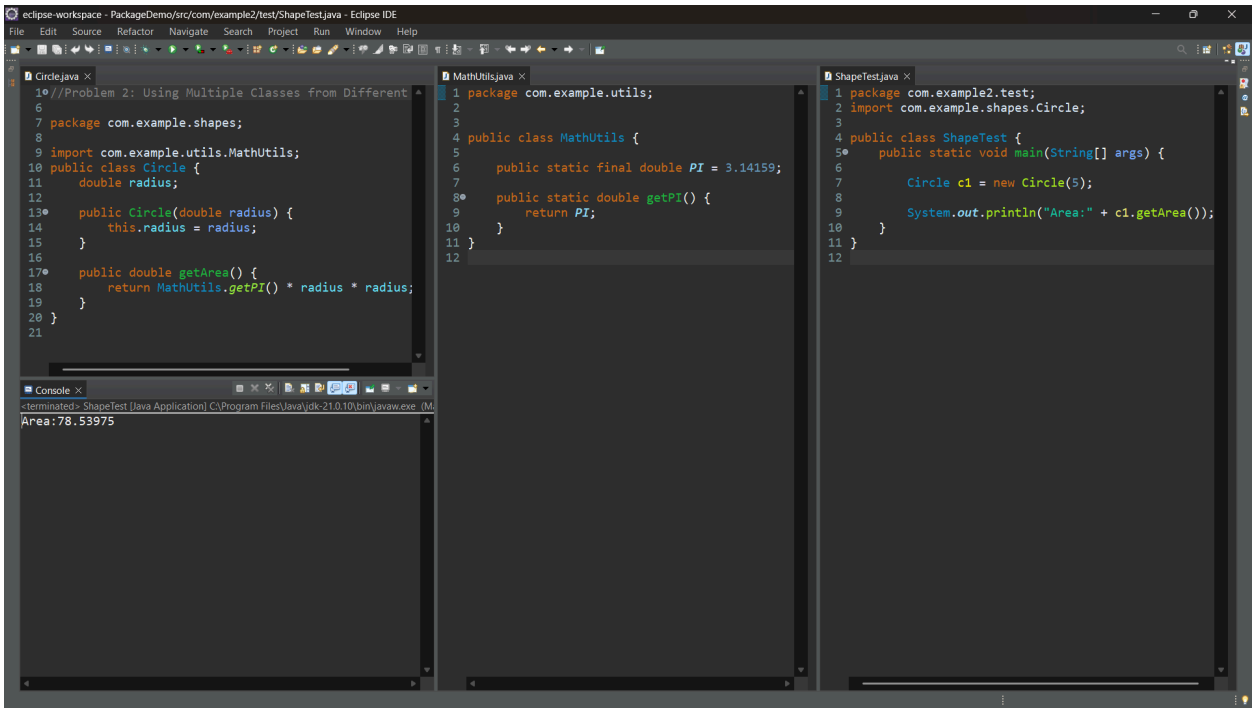
Problem 2: Using Multiple Classes from Different Packages

Task: Create a package com.example.shapes with a class Circle and another package com.example.utils with a class MathUtils. The Circle class should use MathUtils (it should have a method to provide PI value) to calculate the area of the circle. Then, create a ShapeTest class to demonstrate this functionality.

Solution:

```
package com.example.shapes;


import com.example.utils.MathUtils;
public class Circle {
	double radius;


	public Circle(double radius) {
		this.radius = radius;
	}
	public double getArea() {
		return MathUtils.getPI() * radius * radius;
	}
}


package com.example.utils;
public class MathUtils {
	public static final double PI = 3.14159;
	public static double getPI() {
		return PI;
	}
}
```

```
package com.example.utils;

public class MathUtils {

public static final double PI = 3.14159;

        public static double getPI() {

                return PI;

        }

}
```



Problem 3: Creating Sub-Packages

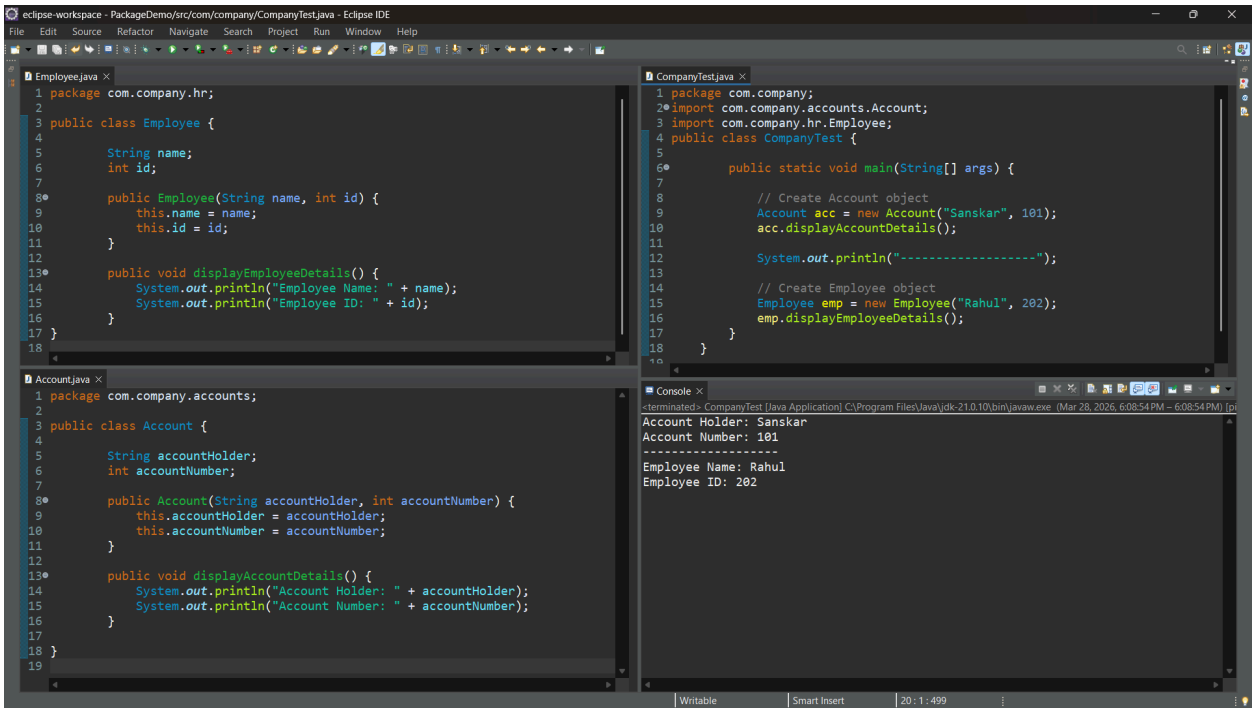
Task: Create a main package com.company with sub-packages accounts and hr. The accounts package should have a class Account with a method displayAccountDetails(), and the hr package should have a class Employee with a method displayEmployeeDetails(). Demonstrate the usage of these classes in a CompanyTest class.

Solution:

```
package com.company.accounts;

public class Account {

    String accountHolder;

    int accountNumber;


    public Account(String accountHolder, int accountNumber) {

        this.accountHolder = accountHolder;

        this.accountNumber = accountNumber;

    }

    public void displayAccountDetails() {

        System.out.println("Account Holder: " + accountHolder);

        System.out.println("Account Number: " + accountNumber);

    }

}


package com.company.hr;

public class Employee {


    String name;

    int id;


    public Employee(String name, int id) {
```

```
        this.name = name;
        this.id = id;
    }

    public void displayEmployeeDetails() {
        System.out.println("Employee Name: " + name);
        System.out.println("Employee ID: " + id);
    }
}

package com.company;
import com.company.accounts.Account;
import com.company.hr.Employee;
public class CompanyTest {
    public static void main(String[] args) {

        // Create Account object
        Account acc = new Account("Sanskar", 101);
        acc.displayAccountDetails();

        System.out.println("-------------------");

        // Create Employee object
        Employee emp = new Employee("Rahul", 202);
```

```
        emp.displayEmployeeDetails();
    }
}
```

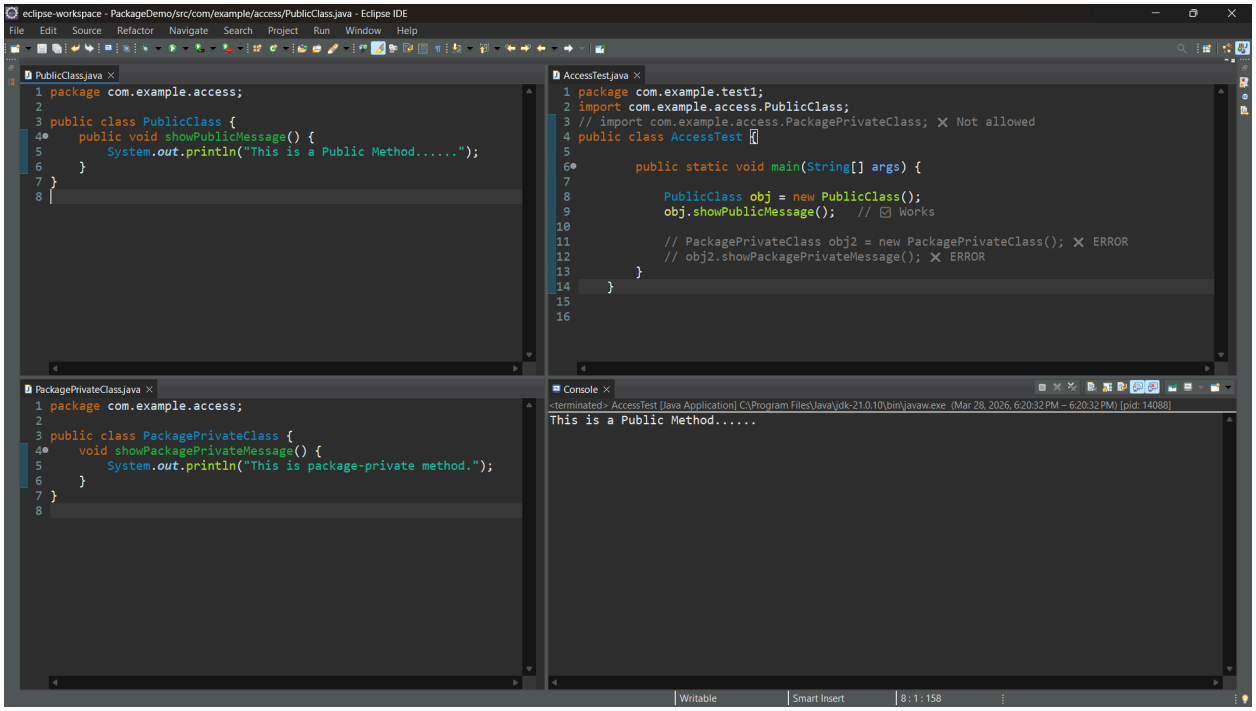


Problem 4: Package-Private Access and Public Classes

Task: Create a package com.example.access with two classes: PublicClass and PackagePrivateClass. The PublicClass should have a public method showPublicMessage(), and the PackagePrivateClass should have a package-private method showPackagePrivateMessage(). Create a class AccessTest to demonstrate that the package-private method cannot be accessed outside its package.

Solution:

```
package com.example.access;
```

```
public class PublicClass {

    public void showPublicMessage() {
        System.out.println("This is a public method");
    }
}

package com.example.access;
class PackagePrivateClass {

    void showPackagePrivateMessage() {
        System.out.println("This is package-private method");
    }
}

package com.example.test;

import com.example.access.PublicClass;
// import com.example.access.PackagePrivateClass;  Not allowed

public class AccessTest {
    public static void main(String[] args) {

        PublicClass obj = new PublicClass();
```

```
        obj.showPublicMessage();   // Works


        // PackagePrivateClass obj2 = new PackagePrivateClass();  ERROR

        // obj2.showPackagePrivateMessage();  ERROR

    }

}
```